

Lynbrook Math Summer Camp 2026

A3 Word Problems

Lynbrook Math Club

August 2026

A3.1 Lecture

Introduction

Have you ever wondered whether your test scores increase the class average? How much money you actually save from discounts? How long you can procrastinate before you actually have to start your homework? Algebra can solve all these problems, and more.

Setting Up Variables

It is hard to write equations with apples and oranges. That is why we use variables to represent unknowns. When defining variables, we try to focus on what is relevant to solving the problem. We also try to choose a variable that is connected to the value it represents. For example, a for the number of apples or o for the number of oranges.

We will now go over common types of word problems.

Conversion Problems

Example 1

In a world that's not so far away, one orange is worth six sporanges, and one sporange is worth five lozenges. How many lozenges is three oranges worth?

Let a lozenge be worth l . Then, a sporange is worth $5l$ and an orange is worth $6(5l) = 30l$. Three oranges would then be worth $3(30l) = 90$ lozenges.

Tips

For more complex conversion problems, consider setting each unit as a separate variable and explicitly writing out equations relating them, such as $F = \frac{9}{5}C + 32$ for converting from Fahrenheit to Celsius. Also remember that when converting rates, like miles per hour, multiply the original rate by a conversion factor equal to one, such as $\frac{1\text{hr}}{60\text{min}}$. This can also be written as $\frac{60\text{min}}{1\text{hr}}$ depending on what the problem calls for.

Percent/Ratio Problems

Example 2

Nitin and Abhigyan go shopping for calculators together. Nitin gets the TI-84, and Abhigyan gets the cheaper CASIO Fx-991EX. The calculators are expensive, so they both use their coupons. Nitin's coupon is 50% off, and Abhigyan's is 25% off. If the price difference between the two calculators drops from \$76 to \$32 after the discounts are applied, how much was the TI-84 originally?

We let the original price of the TI-84 be n , and the original price of the CASIO Fx-991EX be a . The original price difference between the two is \$76, so $n - a = 76$. After the price reductions, the equation becomes $(1 - 50\%)n - (1 - 25\%)a = 32 \Rightarrow 0.5n - 0.75a = 32$. Note that when applying a discount of $x\%$, we need to multiply the original price by $1 - \frac{x}{100}$, thereby removing $x\%$ from the original price.

We now solve the system of equations for n . We solve the first equation for a and plug it into the second equation:

$$\begin{aligned} a &= n - 76 \\ 0.5n - 0.75(n - 76) &= 32 \\ -0.25n + 57 &= 32 \\ 0.25n &= 25 \\ n &= 100 \end{aligned}$$

Therefore, the TI-84 was originally \$100.

Note that we did not need to solve for the price of Abhigyan's calculator in order to solve this problem.

Example 3

Sohum has a 729 page AoPS textbook. At the end of every day, he reads $\frac{1}{3}$ of the pages he has not read yet. At the beginning of Monday, he had not started reading yet. By the end of Friday, how many pages has he not read yet?

Every day, Sohum reads $\frac{1}{3}$ of the pages that he has not read yet, which means he has not read $1 - \frac{1}{3} = \frac{2}{3}$ of the remaining pages each day. From Monday to Friday, five days have passed. This means that by the end of these five days, he will have $729 \cdot (\frac{2}{3})^5 = 96$ pages left to read.

Tips

$X\%$ can be represented by $\frac{X}{100}$ in equations. A discount of $X\%$ off equates to multiplying the original by $\frac{100-X}{100}$. Adding a percentage $X\%$ equates to multiplying the original by $1 + \frac{X}{100}$. This also applies to other problems when a percentage or a fraction of something is removed, such as with teams or food. When applying multiple percentages, note that they can be applied in any order, as finding a percent of something is multiplication, which is commutative.

Comparison Problems

Example 4

In the first year when both George and Yutong made AIME, Yutong was twice George's age. Two years later, both George and Yutong's ages are perfect squares. Assuming both of them are still minors, how old was George when he first made AIME?

Let Yutong's age be y and George's age be g when they both first made AIME. We know that $y = 2g$. We also know that $g + 2$ and $y + 2$ are perfect squares, and that both are less than 18. Since $y = 2g$, y must be even, and $y + 2$ must also be even. Therefore $y + 2$ must be equal to 4 or 16. However, if $y + 2 = 4$, $y = 2$ and $g = 1$, which is impossible because $1 + 2$ is not a perfect square. Therefore $y + 2 = 16$, and $y = 14$. Then $g = \frac{14}{2} = 7$, and George was 7 when he first made AIME (yes, this based off of a true story).

Tips

Comparison problems can usually be solved through systems of equations. If the question only asks for the value of one variable, only solve for that variable without solving for the other variable if possible.

Rate/Distance Problems

Example 5

Benjamin Zhang and Benjamin Oh are racing each other from Rainbow Park to Lynbrook. Benjamin Oh runs at a constant rate of 8 miles per hour, and Benjamin Zhang runs at a constant rate of 10 miles per hour. However, Benjamin Zhang is pushing a marimba along with him, which slows his rate down by 60%. If the distance the two travel is 2000 feet, how much earlier will Benjamin Oh arrive at Lynbrook than Benjamin Zhang, to the nearest second?

Benjamin Zhang's rate is slowed down by 60%, so his new speed is 40% of his original speed, or $0.4(10) = 4$ miles per hour. We notice that this is half of Benjamin Oh's speed, so it will take Benjamin Zhang twice as long to get to Lynbrook. This means that the difference between Benjamin Zhang's time and Benjamin Oh's time will be exactly Benjamin Oh's time. We thus calculate this time.

Since the distance traveled is measured in feet and the rates are measured in miles per hour, we must convert the units of distance so they match. We can either convert feet to miles or miles to feet. We decide to convert from miles to feet to make calculations easier, and also convert from hours to minutes for the same reason. We first convert Benjamin Oh's speed to feet per minute:

$$8\text{mi/hr} \times \frac{5280\text{ft}}{1\text{mi}} \times \frac{1\text{hr}}{60\text{min}} = 704\text{ft/min}.$$

We then find how long it takes Benjamin Oh to run 2000 ft:

$$\frac{2000\text{ft}}{704\text{ft/min}} = \frac{125}{44}\text{min}.$$

We then convert this to seconds:

$$\frac{125}{55}\text{min} \times \frac{60\text{sec}}{1\text{min}} = \frac{1875}{11}\text{sec} = 170\frac{5}{11}\text{sec} \approx 170\text{seconds}.$$

Therefore Benjamin Oh arrives at Lynbrook approximately 170 seconds ahead of Benjamin Zhang.

Example 6

Yutong and Jasmine must write example problems for the LMSC. Their lecture is due in exactly 4 hours, and they must write 16 problems together. Each problem is an average of 500 words. Yutong can write at an average of 15 words per minute, and Jasmine can write an average of 18 words per minute. Assuming they collaborate on writing, will they be able to finish in time?

We first find the total number of words that Yutong and Jasmine must write. 16 problems times 500 words is 8000 words total. Next, we find the combined rate that Yutong and Jasmine write at. Every minute, Yutong writes 15 words and Jasmine writes 18 words, so 33 words total are written every minute. Thus the combined

rate of Yutong and Jasmine is 33 words per minute. We use this rate to find the total time it takes Yutong and Jasmine:

$$\frac{8000\text{words}}{33\text{words/min}} = 242\frac{14}{33} \text{ minutes.}$$

The lecture is due in exactly 4 hours, or 240 minutes. The time it takes Yutong and Jasmine is over 2 minutes longer than that time, so they will not be able to finish the lecture on time.

Tips

The equation $\text{speed} = \frac{\text{distance}}{\text{time}}$ is key. Also remember that for problems where the speed changes over time, the average speed is equal to the total distance over the total time. If two rates are being applied simultaneously to the same thing, as in the previous example problem, the rates can just be added.

Logic Problems

Example 7 (Based off of 1997 PMWC Problems/Problem T5)

Mr. Tsai left a very difficult math problem on the whiteboard. During lunch time, a student solves it anonymously. Trying to figure out who it was, Mr. Tsai picks out the most capable students he knows and questions them. Below are their responses.

Steven: It was either Jessica or Tanish.

Jessica: Catherine and I did not do it, we were together eating lunch.

Tanish: Both Steven and Jessica are lying.

Mason: I believe one of Steven or Jessica is telling the truth.

Catherine: I don't think Mason is telling the truth.

Mr. Tsai knows that two of the students are chronic liars, and the rest always tell the truth. Which student solved the math problem?

We do not know who the two liars are, so we divide the problem into cases.

Case 1: Steven is telling the truth.

If Steven is telling the truth, then Tanish must be lying, since he claims that Steven is lying. If Mason is telling the truth, then only one of Steven or Jessica must be telling the truth, so Jessica must be lying. However, if Mason is telling the truth, Catherine is also lying, which brings the total number of liars to three, which Mr. Tsai knows is not the case. Therefore, Mason is lying, and both Jessica and Catherine are telling the truth. In this case, we know from Steven that either Jessica or Tanish did it, but we know from Jessica that she and Catherine did not do it, so the one who solved the math problem must have been Tanish.

Case 2: Steven is lying.

If Steven is lying, then we know that there must only be one other liar. Catherine's statement directly contradicts Mason's, so one of them must be the other liar. Therefore, Jessica and Tanish must both be telling the truth. However, if Tanish's statement is true, then Jessica is lying. Therefore, there is a contradiction, and Steven cannot be lying.

Therefore, the one who solved the math problem was Tanish.

Tips

Look for contradictions, as they cannot both be true, but they cannot both be false. Also, casework is key. If you are not sure if a statement is true or false, try seeing how the other statements hold up if you set it as either true or false.

Statistics Problems

Problems involving basic statistical concepts are very common in competition. Therefore, we must understand the definitions of these concepts in order to do well.

Definition A3.1.8 (Mean). The mean of a data set is the average of all the numbers of the set. This can be found by adding all the numbers in the set and dividing by the total number of numbers within the set. That is, for a given set $S = [x_1, x_2, x_3, \dots, x_{n-1}, x_n]$ (S has a total of n terms), the average of S is $\frac{x_1 + x_2 + x_3 + \dots + x_{n-1} + x_n}{n}$.

Definition A3.1.9 (Median). Order the elements of a data set from highest to lowest. The median is the element in the exact middle of that list. If there are an even number of elements, the median is the average of the two elements in the middle.

Definition A3.1.10 (Mode). Given a data set, some of the elements within that data set may appear multiple times. The mode is the element that appears the most number of times. There may be multiple modes. If all the elements only appear once, the mode is all of the elements.

Definition A3.1.11 (Range). Given a data set, the range of the data set is the difference of the largest element and the smallest element of the set.

Example 12

All of the ten math club officers qualified for the AIME. However, Steven, and Mason have to take the USABO on the same day as AIME I, so they take the AIME II instead. After the AIME I, the average score amongst the eight math club officers is 13. After the AIME II, the average score dropped down to 12.4. If Mason scored an 8 because he was sick, what was Steven's AIME score?

The average of the eight officer's AIME I scores is the sum of all the officer's scores over eight. Since this is equal to 13, the sum of the eight officer's AIME I scores must be $13 \cdot 8 = 104$. We then set up the equation for the average AIME scores of the entire officer team, given that Mason's score is 8 and Steven's score is s .

$$\frac{104 + 8 + s}{10} = 12.4$$

We then simplify:

$$112 + s = 124 \Rightarrow s = 12$$

Therefore Steven's AIME score is 12.

Tips

For questions with multiple unknowns, list each one as a variable. For problems where the mode is a variable, you can list the same variable multiple times. Typically, equations with averages can be simplified, and then solved with other information given in the problem.